

10,000

THE NUMBER OF GALAXIES  
VISIBLE IN THE HUBBLE  
ULTRA DEEP FIELD

Part of the Hubble Ultra Deep Field, showing some of the farthest galaxies ever seen. The entire image contains about 10,000 individual galaxies, some of which are seen by light that left them 13 billion years ago.

**IN MARCH, NASA RELEASED A REMARKABLE IMAGE** of a tiny part of the sky taken by the Hubble Space Telescope (HST): the **Hubble Ultra Deep Field**, a follow-up to the Hubble Deep Field of 1995. This new image was the result of 800 long exposures—with a total exposure

cranium has a volume about a quarter that of a modern human

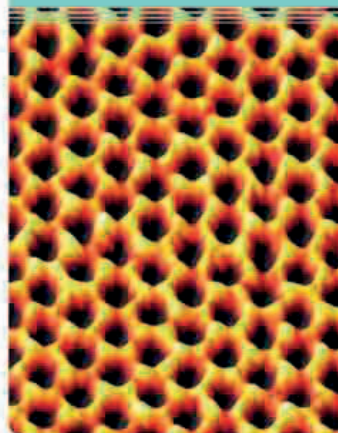
time of nearly 280 hours—captured by an instrument called the Advanced Camera for Surveys, which was installed on the HST in 2002. The image showed thousands of faint galaxies, many of which are so far away that their light would have left them billions of years

ago, when the Universe was very young. Astronomers are still studying the image, which provides new **information about galaxy formation**. In 2012, the HST produced an even more detailed view of the same part of the sky. Called the eXtreme Deep Field, the 2012 image revealed about 5,500 galaxies more than shown in the Ultra Deep Field.

In October, Australian palaeoanthropologists unveiled the partial **remains of an unusual humanlike**

**skeleton**: a 3ft- (1m-) tall adult with a small skull that would have housed a very small brain. Australian and Indonesian archaeologists discovered the specimen in a cave on the

#### GRAPHENE



Graphene takes its name from graphite, which is composed of layers of carbon atoms. In graphite, the layers move over each other easily, which is why it is slippery, but they are extremely robust because the atoms form strong hexagonal bonds, as shown in this color-enhanced electron micrograph; graphene is simply a single layer of graphite.

Island of Flores, east of Bali, Indonesia, together with sophisticated tools and the remains of animals. The skeletal remains were around 18,000 years old, and the researchers confirmed them as being from a **new species in the genus *Homo***, which includes our closest ancestors, such as *Homo erectus*, as well as our own species, *Homo sapiens*. The researchers called the **new species *Homo floresiensis***.

Remains of a total of seven individuals were found altogether. The indigenous people of Flores have ancient but detailed legends about a race of small, hairy people who murmured in their own language, and so it is likely that *Homo floresiensis* lived side-by-side with modern humans.

Also in October, scientists at Manchester University, UK, succeeded in producing **samples of graphene** (see panel, above), a form of pure carbon not previously produced in bulk. The new material was just one atom thick but remarkably strong, transparent, and was later found to conduct electricity at room temperature.

**“ THIS CANNOT BE A PECULIAR MODERN HUMAN. ”**

Chris Stringer, British anthropologist, 2004

#### *Homo floresiensis* skull

This skull of the recently extinct, newly discovered species of human, *Homo floresiensis*, is from a female who would have been about 3ft (1m) tall. The small, human features led to this species being nicknamed the “Hobbit.”

maxilla  
(upper jaw)

mandible  
(lower jaw)

March 9 NASA publishes the Hubble Ultra Deep Field

May 17 Stockholm Convention on Persistent Organic Pollutants comes into force, banning certain environmentally damaging chemicals

June 22 US tetraplegic patient Matthew Nagle becomes the first person to have a brain-computer interface, enabling him to control a computer by thought

June 21 SpaceShipOne becomes the first private spacecraft to reach space

September Launch of the Supernova Early Warning System, for detecting neutrino bursts that signal a supernova explosion

October 22 Russian-born physicists Andre Geim and Kostya Novoselov create sheets of graphene

October 28 18,000-year-old fossil remains of a close human relation, *Homo floresiensis*, unveiled by paleontologists